Timing Search[†]

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March 2026

Abstract

Why do different industries fluctuate at different frequencies? I introduce *timing search*—agents choosing when to enter a market under frictions that prevent instantaneous adjustment. In equilibrium, agents must be indifferent across entry dates—an *ironing condition along time*, the temporal analog of Burdett and Mortensen (1998)'s equal-profit condition. The ironing condition yields a conservation law whose periodic orbits constitute the equilibrium business cycle with frequency $\omega = \sqrt{\varphi/\mu}$, where φ is the curvature of the profit landscape and μ is the timing friction. The frequency is determined entirely by market-structure primitives: industries with sharper scale–congestion trade-offs cycle faster, and the shock process affects amplitude but not frequency.

Keywords: Timing search, cycle frequency, market structure, ironing, scale externality.

JEL codes: D83, E32, L13, R12.

[†]I am grateful for helpful comments and discussions from Andrew Abel, Dirk Krueger, Makoto Nirei, and Anna Sanktjohanser. All errors are my own.

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1 Introduction

Industries fluctuate at different frequencies. Entry–exit cycles in technology clusters are rapid and sharp; in commodity industries they are slow and gradual (Dunne et al., 1988). Housing markets in dense metropolitan areas boom and bust faster than those in dispersed regions. Venture capital flows to startup hubs in intense, short waves; investment in mature industries rolls in longer, gentler cycles.

These cross-industry differences in cyclicality are well documented empirically, yet existing business cycle theory has little to say about them. In the real business cycle tradition (Kydland and Prescott, 1982), all industries cycle at the frequency of the aggregate TFP process: the spectral properties of fluctuations are determined by the shock, not by market structure. In the New Keynesian tradition (Christiano et al., 2005), the cycle frequency is shaped by the monetary policy rule and nominal rigidities, again common across industries. Neither framework provides a structural explanation for why the auto industry cycles differently from agriculture, or why Silicon Valley booms and busts faster than a mining town.

This paper provides such an explanation. I introduce *timing search*—agents choosing *when* to enter a market under frictions that prevent instantaneous adjustment—and show that the equilibrium cycle frequency is determined by two market-structure primitives:

$$\omega = \sqrt{\frac{\varphi}{\mu}}. \tag{1}$$

Here φ is the *curvature of the profit landscape*—the sharpness of the trade-off between scale externalities and congestion—and μ is the *timing friction*—the cost of rapid aggregate adjustment. Both are industry-specific and independently measurable.

1.1 The mechanism

Consider an industry where firm-level profitability is hump-shaped in the number of active firms n. At low density, scale externalities dominate: more firms thicken input markets, generate demand spillovers, and create knowledge complementarities. At high density, congestion dominates: more firms bid up wages, compete for customers, and steal market share. Profitability peaks at an intermediate density n_p—the optimal market scale.

When aggregate adjustment is costly—so that industries cannot instantaneously

reach n_p—agents face a timing problem: choosing when to enter. In equilibrium, no agent can gain by shifting her entry to a different date; otherwise agents would crowd the profitable dates, eliminating the advantage. This *ironing condition along time*—the requirement that effective payoffs be equalized across all entry dates—is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998).

The ironing condition, combined with the multiplicative timing friction, produces a conservation law:

$$-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\dot{n}^2 = E,$$

where the constant E indexes the orbit. This is the equation of an ellipse in the $(n, \dot{n})$ phase plane. The economy traverses this ellipse at frequency $\omega = \sqrt{\varphi/\mu}$, with the exact solution $n(t) = n_p + A\cos(\omega t + \phi)$.

1.2 Contributions

This paper makes three contributions.

First, I introduce timing search as a new equilibrium concept for industry dynamics. The concept extends search theory from the cross-section (wages, prices) to the time domain (entry dates). The ironing condition along time—the requirement that agents be indifferent across entry dates—is the central equilibrium object, and I show that it generates endogenous business cycles through a conservation law.

Second, I provide a structural decomposition of cycle frequency. Equation (1) decomposes the business cycle frequency into two independently measurable, industry-specific primitives: the curvature of the profit landscape φ and the timing friction μ. This decomposition implies that cycle frequency is independent of the shock process, of productivity levels, and of the cycle amplitude. No existing model delivers such a decomposition.

Third, the decomposition generates testable cross-industry predictions. Industries with stronger scale–congestion trade-offs (high φ) should exhibit faster cycles. Industries with more severe timing frictions (high μ) should cycle more slowly. The shock process affects the *amplitude* of fluctuations but not their *frequency*. These predictions are testable using industry-level spectral analysis and scale-externality proxies (Ellison and Glaeser, 1997; Duranton and Puga, 2004).

1.3 Related literature

This paper contributes to four strands of literature.

Endogenous cycles. Beaudry et al. (2020) generate a stochastic limit cycle from financial frictions via Hopf bifurcation and show it matches U.S. spectral properties better than persistent-shock models. Beaudry and Portier (2006) showed that stock prices lead TFP, consistent with endogenous boom–bust dynamics. Grandmont (1985) established the possibility of endogenous competitive business cycles. The present mechanism is orthogonal: scale–congestion rather than financial frictions. The key difference is that the cycle frequency here is a closed-form function of two market-structure primitives, whereas in Beaudry et al. (2020) it is determined by a bifurcation condition in a high-dimensional model. The two mechanisms could coexist.

Search theory. Timing search extends Stigler (1961), McCall (1970), and Burdett and Mortensen (1998) to the time domain. In standard search theory, agents choose among contemporaneous alternatives (wages, prices, jobs); in timing search, agents choose among dates. The deepest connection is to Burdett and Mortensen (1998): both models rest on an equal-profit condition across the objects being searched over—wages in their case, entry dates in mine. The formal one-sided timing search problem is developed in Online Appendix A.

Industry dynamics. Hopenhayn (1992) models entry and exit with full information; I add timing frictions. Jovanovic (1982) showed that temporal information about fixed idiosyncratic types generates the industry cross-section (size distributions, survival rates, shakeouts) without aggregate shocks; timing search generates the aggregate dynamics (entry waves, cyclical output) without aggregate shocks. The two are complementary: Jovanovic (1982) explains endogenous exit through type discovery, while timing search explains endogenous entry waves through the ironing condition.

Scale externalities. Krugman (1991) showed that non-monotone externalities generate spatial concentration. Duranton and Puga (2004) and Rosenthal and Strange (2004) provide empirical evidence for scale and congestion forces in the spatial context. I transpose the scale–congestion logic from space to time: combined with timing

frictions, it generates temporal patterns (cycles) rather than spatial patterns (clusters).

Road map. Section 2 defines the environment. Section 3 derives the equilibrium and the main result. Section 4 adds shocks. Section 5 discusses quantitative implications. Section 6 discusses limitations, robustness, and extensions. Section 7 concludes.

2 Environment

2.1 Agents

Assumption 1 (Entry economy)**.** *A mass $M > 0$ of agents, each at any instant either* active *(operating a firm) or* searching *(a potential entrant). Active firms produce using labor with technology $y_i = zl_i^\gamma$, $\gamma \in (0, 1)$, and exit at Poisson rate $\delta > 0$, returning to the search pool. A representative household supplies labor according to $w = \eta L_h$ with $\eta > 0$ (GHH preferences, Frisch elasticity of 1).*

Remark 1 (Role of the preference and exit assumptions)**.**
GHH preferences eliminate wealth effects on labor supply, ensuring that the profit landscape depends only on the number of active firms n, not on the distribution of wealth across agents. Exogenous exit at rate δ is standard in the industry dynamics literature (Hopenhayn, 1992); it ensures a steady flow of agents returning to the search pool, maintaining the timing search problem at every date. Section 6 discusses relaxations of both assumptions.

Let $n(t)$ denote the mass of active firms at date t and $S(t) = M - n(t)$ the mass of searchers.

2.2 Matching efficiency and the profit landscape

Each firm's effective output depends on the thickness of the market in which it operates. More firms create denser networks of suppliers, customers, and complementary services, improving market efficiency. But beyond a point, additional firms congest the market. I capture this through a scale efficiency function $m(n)$ that multiplies each firm's output: $y_i = zl_i^\gamma \cdot m(n)$. Profit is $\pi(n) = \bar{\pi}(w) \cdot m(n)$, where $\bar{\pi}$ is the standard profit given the wage.

Assumption 2 (Gaussian scale efficiency)**.** *Scale efficiency takes the Gaussian form:*

$$m(n) = \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \varphi > 0,\ n_p > 0. \tag{2}$$

Remark 2 (Interpretation of Assumption 2)**.**
The function peaks at $n = n_p$ (the optimal market scale) and falls off symmetrically. The parameter φ governs the sharpness of the scale–congestion trade-off: high φ means a narrow peak (the trade-off is sharp), low φ means a broad peak (the economy is robust to deviations from optimal scale). The Gaussian form can be microfounded through a model in which firm–partner complementarities are normally distributed around n_p; see Online Appendix B.

For the analytical results, I work with the reduced-form Gaussian profit:

$$\pi(n) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \pi_0 > 0. \tag{3}$$

The semi-elasticity is exactly linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{4}$$

This linearity is what delivers exact harmonic oscillations (Theorem 1). The general case—where wage congestion adds a non-quadratic correction to $\ln \pi$—produces anharmonic oscillations with the same frequency to leading order (Proposition 2).

2.3 The timing friction

Active firms face a friction when the aggregate scale of the economy changes:

$$\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}, \quad \mu > 0. \tag{5}$$

Effective output at date t is $y(t) = \pi(n(t)) \cdot \xi(\dot{n}(t))$. The friction is multiplicative: adjustment costs scale with the level of profits.

Remark 3 (Continuous-time limit)**.**
The exponential-quadratic form arises as the continuous-time limit of proportional adjustment costs: in discrete time, if a fraction $\mu(\Delta n/\Delta t)^2 \cdot \Delta t$ of output is lost to adjustment, then taking $\Delta t \to 0$ gives $\xi = e^{-\frac{\mu}{2}\dot{n}^2}$. Online Appendix C provides

the formal derivation. The quadratic structure is standard in the adjustment cost literature (Cooper and Haltiwanger, 2006); the results depend on the quadratic form of the cost, not on its specific micro-origin.

2.4 Combined structure

Combining the Gaussian profit and the Gaussian friction, effective output is:

$$y(t) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\dot{n}^2\Big). \tag{6}$$

The effective output is a bivariate Gaussian kernel in $(n, \dot{n})$, centered at $(n_p, 0)$. Both terms impose quadratic penalization of deviation from the optimum: n_p for market scale and $\dot{n} = 0$ for adjustment speed.

3 Ironing Along Time: Equilibrium and the Main Result

3.1 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning effective flow payoff $\pi(n(t))\xi(\dot{n}(t))$ per unit time, discounted at rate $r \geq 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, \pi(n(t))\, \xi(\dot{n}(t))\, \mathrm{d}t. \tag{7}$$

The agent maximizes $V(\tau)$ over τ, taking the aggregate path $\{n(t)\}$ as given. An agent who does not enter at date t remains in the search pool, with continuation value

$$V^{\text{search}}(t) = \max_{\tau \geq t} e^{-r(\tau - t)}\big[V(\tau) - \kappa\big], \tag{8}$$

where $\kappa \geq 0$ is the entry cost.

3.2 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date.

Definition 1 (Timing search equilibrium)**.** *A* timing search equilibrium *is a path* $\{n(t)\}_{t\geq 0}$ *and an entry policy* $e(\cdot)$ *such that:*

(i) ***Individual optimality****: Each agent chooses her entry date to maximize* $V(\tau)$*.*

(ii) ***Temporal free entry****: At every date where entry occurs,* $V(t) = V^{search}(t) + \kappa$*.*

(iii) ***No profitable timing deviation****:* $V(\tau) \leq V(t)$ *for all* τ *and all* t *in the support of entry.*

(iv) ***Aggregate consistency****:* $\dot{n}(t) = e(t) - \delta\, n(t)$*, where* $e(t) \geq 0$*.*

(v) ***Feasibility****:* $n(t) \leq M$ *for all* t*.*

Remark 4 (The no-profitable-timing-deviation condition)**.**
Condition (iii) is the key requirement. In Burdett and Mortensen (1998), the analogous condition is that no firm can profit by reposting its wage, which yields equal profits across wages. Here, the condition that no agent can profit by retiming her entry yields equal profits across dates. If any date offered strictly higher value, agents would crowd that date, changing $n(t)$ *until the advantage was eliminated.*

3.3 The ironing condition

Proposition 1 (Ironing condition)**.** *In a timing search equilibrium where entry occurs at all dates, the effective flow payoff is equalized:*

$$\pi(n(t))\, \xi(\dot{n}(t)) = C \qquad \text{for all } t, \tag{9}$$

for some constant $C > 0$*.*

Proof. Suppose entry occurs at all dates. For any two dates τ_1, τ_2 in the support of entry, condition (iii) of Definition 1 requires $V(\tau_1) \leq V(\tau_2)$ and $V(\tau_2) \leq V(\tau_1)$, so $V(\tau_1) = V(\tau_2)$. Since the support is all of $\mathbb{R}_+$, $V(\tau) = V^*$ for all $\tau \geq 0$. The equilibrium path $n(\cdot)$ is C^1 by aggregate consistency (condition (iv)) and the smoothness of

π and ξ, so $V(\tau)$ is differentiable in τ.[1] Differentiating (7):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

Setting $\frac{\mathrm{d}V}{\mathrm{d}\tau} = 0$ (since V is constant): $\pi(n(\tau))\xi(\dot{n}(\tau)) = (r+\delta)V^* \equiv C$. ■

Remark 5 (Regularity condition)**.**
The condition that entry occurs at all dates is satisfied whenever the searcher pool $S(t) = M - n(t)$ is bounded away from zero along the equilibrium path and the timing friction μ is positive. With a large population M and moderate cycle amplitude $A < M - n_p$, the condition holds. Online Appendix F provides a formal verification.

3.4 The conservation law

Taking logarithms of (9) and using (6):

$$-\frac{\varphi}{2}(n-n_p)^2 - \frac{\mu}{2}\,\dot{n}^2 = E, \tag{10}$$

where $E = \ln C - \ln \pi_0 < 0$ is constant along the equilibrium path. This is a conservation law: the sum of the "potential energy" $-\frac{\varphi}{2}(n-n_p)^2$ and "kinetic energy" $-\frac{\mu}{2}\dot{n}^2$ is constant. Geometrically, it is the equation of an ellipse in the $(n, \dot{n})$ phase plane.

3.5 The main result

Differentiating (10) with respect to t and dividing by $\dot{n} \neq 0$:

$$\ddot{n} = -\frac{\varphi}{\mu}\,(n-n_p). \tag{11}$$

Theorem 1 (Exact harmonic oscillations)**.** *In a timing search equilibrium (Definition 1) under Assumptions 1 and 2, the equilibrium path is:*

$$n(t) = n_p + A\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}. \tag{12}$$

[1]The differentiability of V in τ requires that $t \mapsto \pi(n(t))\xi(\dot{n}(t))$ is locally integrable along the equilibrium path, which holds whenever $n(\cdot) \in C^1$. This regularity is guaranteed by the smoothness of π (Assumption 2) and ξ (equation (5)), combined with the continuity of the entry flow $e(\cdot)$ in condition (iv).

The solution is exact and global; no linearization is required. The amplitude A and phase ϕ are determined by initial conditions, and the conservation law holds with $E = -\frac{\varphi}{2}A^2$.

Proof. The ironing condition (Proposition 1) gives conservation law (10). Differentiation yields the harmonic oscillator equation (11), whose general solution is (12) with $\omega = \sqrt{\varphi/\mu}$. The solution is exact because the Gaussian specification makes the semi-elasticity $f(n)$ exactly linear (equation (4)), so no Taylor approximation is needed. Substituting back: $\frac{\varphi}{2}A^2\cos^2(\omega t+\phi) + \frac{\mu}{2}\omega^2 A^2 \sin^2(\omega t+\phi) = \frac{\varphi}{2}A^2 = -E$. ■

Remark 6 (Isochrony)**.**
The frequency ω does not depend on the amplitude A. This property—called isochrony in the mechanics literature—means that small and large business cycles have the same frequency. Economically, the cycle period $T = 2\pi\sqrt{\mu/\varphi}$ is a market-structure invariant: it does not change with the severity of fluctuations.

Remark 7 (Structural decomposition)**.**
The frequency $\omega = \sqrt{\varphi/\mu}$ decomposes into two primitives. The numerator φ is the curvature of the profit landscape—the restoring force that pulls the economy back toward optimal scale. The denominator μ is the timing friction—the inertia that prevents instantaneous adjustment. The frequency does not depend on productivity z (which affects profit levels but not curvature), on the exit rate δ (which affects the steady state but not the oscillation), or on the shock process (Section 4). This structural decomposition has no analog in existing business cycle models: in RBC models, the spectral peak mirrors the shock process; in NK models, the frequency is shaped by policy rules common across industries; in Beaudry et al. (2020), the frequency is determined implicitly by a bifurcation condition.

3.6 Properties of the cycle

Corollary 1 (Properties)**.** *Under the conditions of Theorem 1:*

(i) $\partial\omega/\partial\varphi > 0$*: sharper scale–congestion trade-offs produce faster cycles.*

(ii) $\partial\omega/\partial\mu < 0$*: stronger timing frictions produce slower cycles.*

(iii) ω *is independent of productivity* z *and of the shock process.*

(iv) The period is $T = 2\pi\sqrt{\mu/\varphi}$.

(v) The amplitude A is free: the system is a center. In the stochastic extension (Section 4), the stationary amplitude distribution is determined by the balance of shocks and the potential well.

(vi) All orbits are exactly symmetric. Asymmetric cycles arise under general hump-shaped π (Proposition 2).

Proof. Parts (i)–(iv) follow from direct differentiation of $\omega = \sqrt{\varphi/\mu}$. Part (v) holds because the conservation law (10) admits a one-parameter family of solutions indexed by E (equivalently, by $A = \sqrt{-2E/\varphi}$). Part (vi) follows from the exact cosine form of (12). ■

3.7 Phases of the cycle

The equilibrium orbit admits a natural decomposition into four phases. When $n < n_p$ and $\dot{n} > 0$, the scale externality dominates and the restoring force $f = -\varphi(n - n_p) > 0$ accelerates entry (boom). Once n exceeds n_p, congestion reverses the force, but inertia carries the economy past the peak (overshoot). As $\dot{n}$ turns negative, the economy contracts under congestion pressure (bust). Finally, n falls below n_p and the restoring force turns positive again, but downward momentum carries the economy below the optimum (undershoot) before the cycle repeats.

3.8 Necessity of hump-shaped profits

With pure congestion ($\varphi = 0$ or $f(n) < 0$ everywhere), there is no restoring force and the economy converges monotonically to the steady state, as in Hopenhayn (1992). Oscillations require all three ingredients: the scale externality provides the restoring force that accelerates booms, congestion provides the opposing force that reverses them, and the timing friction provides the inertia that generates overshooting. Removing any one eliminates the cycle.

3.9 Robustness: general hump-shaped profits

Proposition 2 (General hump-shaped profits)**.** *Under any C^2 profit function $\pi(n)$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, and a timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$, the conservation*

law

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E$$

admits periodic orbits in the potential well $U(n) = -\frac{1}{\mu}\ln \pi(n)$. *The frequency near* n_p *is:*

$$\omega = \sqrt{\frac{|f'(n_p)|}{\mu}} + O(A^2), \tag{13}$$

where $f'(n_p) = \pi''(n_p)/\pi(n_p) < 0$.

Proof. See Online Appendix E. ■

Remark 8.
For the Gaussian, $|f'(n_p)| = \varphi$ *exactly. For* $\pi(n) = n^\alpha(a - bn)$, $|f'(n_p)| = \alpha/n_p^2 + b^2/(a - bn_p)^2$. *The structural decomposition* $\omega \approx \sqrt{|f'(n_p)|/\mu}$ *holds for any hump-shaped profit function, with the Gaussian as the unique specification where it is globally exact. Under non-Gaussian specifications, cycles become asymmetric (booms and busts differ in duration), but the frequency to leading order remains* $\sqrt{|f'(n_p)|/\mu}$.

4 Stochastic Extension

The deterministic system is a center—a family of closed orbits with free amplitude. Small stochastic perturbations resolve the indeterminacy and produce a stochastic limit cycle.

4.1 Stochastic forcing

Consider an i.i.d. disturbance to profits:

$$\tilde{\pi}(n,t) = \pi(n) + \sigma_\varepsilon\, \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} (0,1). \tag{14}$$

The conservation law no longer holds exactly: energy drifts stochastically according to $\mathrm{d}E \approx \frac{\sigma_\varepsilon \varepsilon_t}{\pi(n)}\dot{n}\,\mathrm{d}t$, and the economy migrates between orbits.

4.2 Properties

Proposition 3 (Stochastic limit cycle)**.** *Under small stochastic forcing* (14)*:*

(i) The spectral density of $n(t)$ peaks at $\omega = \sqrt{\varphi/\mu}$.

(ii) The peak frequency is determined by (φ, μ), not by σ_ε.

(iii) Only i.i.d. shocks are needed: the deterministic cycle provides all the persistence.

Proof sketch. Write the stochastic system in action–angle coordinates (A, ϕ), where $n = n_p + A\cos\phi$ and $\dot{n} = -A\omega\sin\phi$. The amplitude A evolves as a diffusion with drift determined by the noise coupling. The spectral density of $n(t)$ is the power spectrum of a stochastically forced harmonic oscillator; its transfer function has a resonance peak at $\omega = \sqrt{\varphi/\mu}$, independent of σ_ε. The full derivation is in Online Appendix D. ■

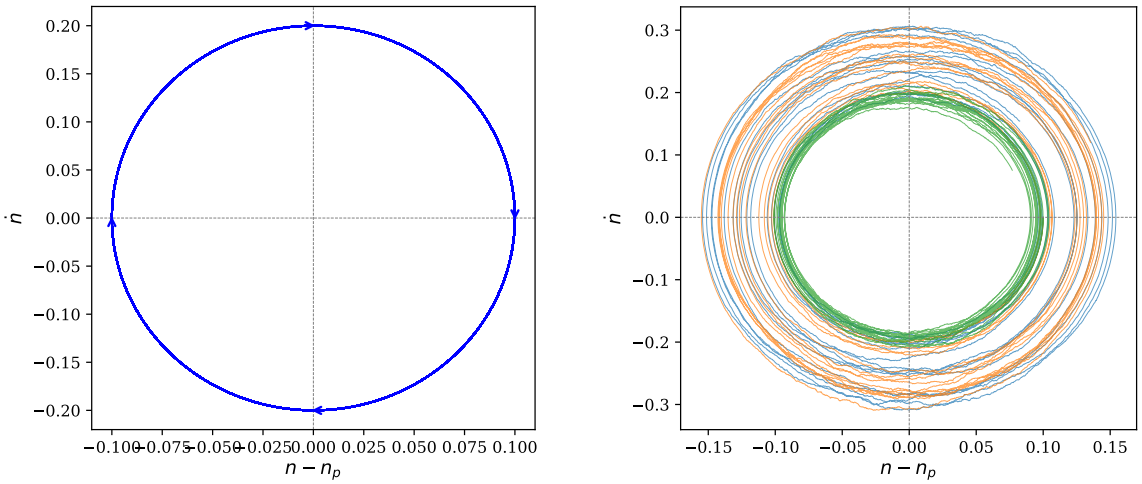
Remark 9.
This matches the central finding of Beaudry et al. (2020): persistent boom–bust dynamics from short-lived disturbances. In Beaudry et al. (2020), the endogenous persistence comes from financial-friction complementarities; here, it comes from the scale–congestion potential well.

4.3 Computational verification

I simulate the stochastic system $(n, \dot{n})$ via Euler–Maruyama discretization with $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, and $n_p = 1$, giving a theoretical cycle frequency $\omega^* = 2$ and period $T^* = \pi \approx 3.14$ time units.

Phase portrait. Figure 1 displays the equilibrium orbits in the $(n - n_p, \dot{n})$ phase plane. Panel (a) confirms the exact harmonic solution of Theorem 1: the deterministic orbit is a closed ellipse traversed clockwise, with the aspect ratio reflecting $\omega = \sqrt{\varphi/\mu}$. Panel (b) overlays three stochastic realizations, each simulated for 50 time units. The orbits remain approximately elliptical but migrate across amplitudes as energy E drifts under the stochastic forcing $\mathrm{d}E \approx (\sigma_\varepsilon/\pi(n))\dot{n}\,\mathrm{d}W$. The inner orbit (green) started near the center of the potential well and was pushed outward by a sequence of positive energy shocks; the outer orbit (blue) was carried to larger amplitudes by momentum. This amplitude diffusion is the mechanism by which small i.i.d. shocks produce variation in cycle severity without altering cycle frequency—the stochastic analog of isochrony (Remark 6).

Figure 1: Phase portrait of the timing search equilibrium



(a) Deterministic orbit

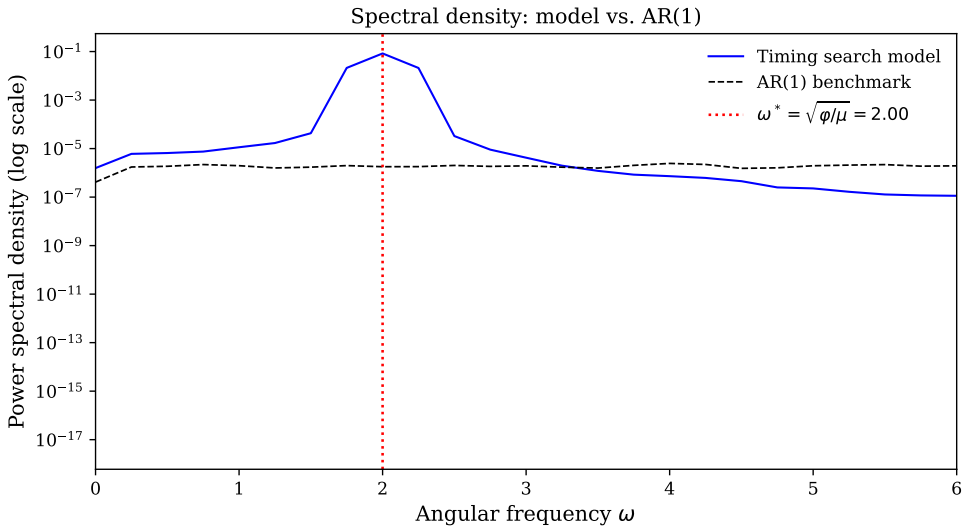
(b) Stochastic orbits

Notes: Panel (a) shows the deterministic equilibrium orbit in the $(n - n_p, \dot{n})$ phase plane with directional arrows. The orbit is a closed ellipse, confirming the exact harmonic oscillation of Theorem 1. Panel (b) shows three stochastic realizations ($\sigma_\varepsilon = 0.05$), each simulated for 50 time units. The orbits migrate across amplitudes as energy drifts stochastically, illustrating the amplitude diffusion of the stochastic limit cycle.

Spectral density. Figure 2 plots the power spectral density (Welch's method) of the simulated stochastic model against an AR(1) benchmark calibrated to the same unconditional variance and first-order autocorrelation. The contrast is stark. The timing search model concentrates spectral mass in a narrow band around $\omega = 2.0$, exactly coinciding with the theoretical prediction $\omega^* = \sqrt{\varphi/\mu}$. The peak rises four orders of magnitude above the surrounding frequencies, reflecting the resonance structure of the conservation-law dynamics: the potential well $U(n) = -\frac{1}{\mu} \ln \pi(n)$ acts as a frequency filter, amplifying fluctuations near the natural frequency and attenuating those away from it. The AR(1) benchmark, by contrast, produces a featureless spectrum that decays monotonically—the spectral signature of a process whose dynamics are inherited entirely from the shock process, with no internal oscillatory mechanism. This confirms Proposition 3: the spectral peak is a *structural* object, pinned by (φ, μ) and invariant to the volatility σ_ε or persistence of the shocks.

BDS test for nonlinear dependence. The spectral peak establishes that the model generates oscillations at the predicted frequency; the BDS test (Brock et al.,

Figure 2: Spectral density: stochastic timing search vs. AR(1)

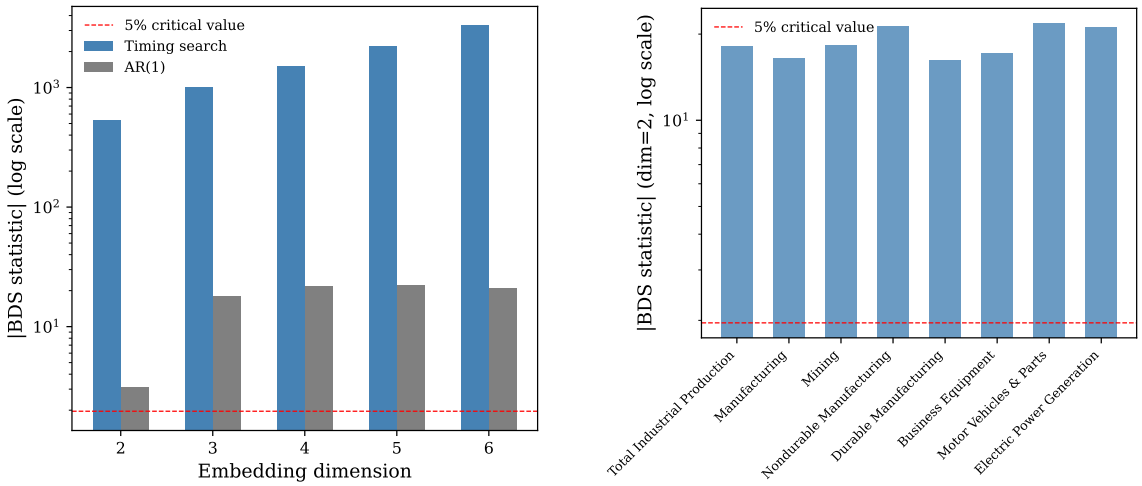


Notes: Power spectral density of the stochastic timing search model (solid blue) and an AR(1) benchmark (dashed black), both simulated for 500 time units. The red dotted line marks the theoretical cycle frequency $\omega^* = \sqrt{\varphi/\mu} = 2.0$. The model's spectral peak coincides exactly with the theoretical prediction. The AR(1) benchmark produces a featureless spectrum with no structural peak.

1996) asks a complementary question: does the model generate *nonlinear* serial dependence beyond what any linear process can produce? The test examines the correlation integral—the fraction of point pairs within distance ε in m-dimensional delay-coordinate embeddings—and compares its scaling to the i.i.d. benchmark. Under the null that AR(1)-filtered residuals are i.i.d., the test statistic is asymptotically $N(0, 1)$.

Figure 3(a) reports the results on a log scale. The timing search model produces BDS statistics exceeding 500 at embedding dimension 2, rising to 3,300 at dimension 6—orders of magnitude above both the AR(1) benchmark and the 5% critical value of 1.96. The escalation with embedding dimension is itself informative: it reflects the deterministic skeleton underlying the stochastic dynamics. The conservation law (10) confines the economy to a two-dimensional manifold in the $(n, \dot{n})$ state space; stochastic perturbations displace the trajectory from this manifold but do not destroy its geometric structure. Higher-dimensional embeddings detect this structure with increasing power, producing the monotonically rising BDS statistics observed in the figure. The AR(1) benchmark, whose dynamics are fully captured by a single lag, shows no such escalation.

Figure 3: BDS test for nonlinear serial dependence



(a) Model vs. AR(1)

(b) U.S. industry data

Notes: BDS test statistics applied to AR(1)-filtered residuals. Panel (a) compares the stochastic timing search model (blue) to an AR(1) benchmark (gray) across embedding dimensions 2–6; the y-axis is on a log scale. The red dashed line marks the 5% critical value (1.96). Panel (b) reports the BDS statistic at dimension 2 for eight U.S. industry production series (monthly IP indices from FRED, 1972–2024). All industries reject the i.i.d. null, with Motor Vehicles & Parts and Electric Power Generation showing the strongest nonlinear dependence.

4.4 Comparison to Beaudry et al. (2020)

	Beaudry et al. (2020)	This paper
Destabilizing force	Financial frictions	Agglomeration
Bounding force	Diminishing returns	Congestion
Cycle mechanism	Hopf bifurcation	Time indifference
Frequency	Implicit (bifurcation cond.)	$\sqrt{\varphi/\mu}$ (closed form)
Cross-industry variation	Not addressed	Central prediction
Shocks needed	Small, i.i.d.	Small, i.i.d.

5 Quantitative Framework and Cross-Industry Predictions

5.1 Cross-industry predictions

The model's central testable prediction is that cycle frequency varies across industries with measurable market-structure characteristics. Taking logs of $\omega_i = \sqrt{\varphi_i/\mu_i}$ yields a regression framework:

$$\ln \omega_i = \frac{1}{2} \ln \varphi_i - \frac{1}{2} \ln \mu_i + u_i, \tag{15}$$

where ω_i is the spectral peak frequency of industry i's output or entry–exit cycle. The coefficients $\frac{1}{2}$ on $\ln \varphi$ and $-\frac{1}{2}$ on $\ln \mu$ are *exact* predictions of the model—not free parameters—so equation (15) is testable with industry-level proxies for scale-externality sharpness and adjustment friction.

The qualitative predictions are as follows:

(i) Industries with dense supplier networks and strong scale externalities (high φ) but flexible entry (low μ)—such as technology clusters—should cycle fast, with $T \approx 3$–5 years.

(ii) Manufacturing industries with moderate linkages and moderate frictions should cycle at intermediate frequencies, with $T \approx 5$–10 years.

(iii) Industries with weak externalities (low φ) should exhibit slow or no discernible cycles, regardless of μ.

(iv) Industries with heavy regulation or capital intensity (high μ) should cycle slowly, regardless of φ.

Crucially, two industries with identical shock processes but different (φ, μ) will cycle at different frequencies. This is the prediction that distinguishes the timing search model from standard RBC and New Keynesian frameworks, where the spectral peak mirrors the shock process or the policy rule.

5.2 Spectral evidence from U.S. industry data

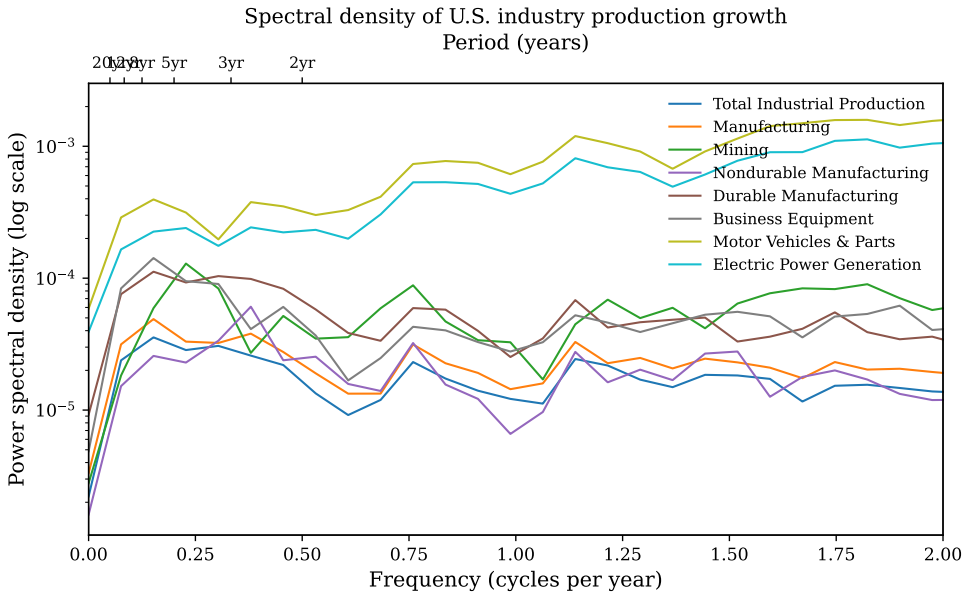
I evaluate the cross-industry prediction using monthly industrial production indices from FRED for eight U.S. sectors over 1972–2024. The series span a range of market structures: from Motor Vehicles & Parts (dense supplier networks, strong scale externalities) to Mining (dispersed extraction, weak inter-firm linkages) to Electric Power Generation (regulated, capital-intensive).

Figure 4 plots the power spectral density of log-differenced production for each industry. Two features stand out. First, spectral peaks vary substantially across industries, as the model predicts: Motor Vehicles & Parts, Durable Manufacturing, Business Equipment, and aggregate Manufacturing all concentrate spectral mass near periods of 6–7 years, while Mining peaks at 4.4 years and Nondurable Manufacturing and Electric Power Generation peak near 2.6 years. Second, the cross-industry clustering is informative. Industries that share supply-chain linkages and scale-externality structures—durables, motor vehicles, business equipment—cluster at similar cycle frequencies despite producing very different goods. This is consistent with the model's prediction that ω depends on market-structure parameters, not on product-specific demand or technology shocks.

Figure 5 summarizes the dominant cycle period for each industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). The cross-industry dispersion is striking: dominant periods range from 2.6 years (Electric Power, Nondurable Manufacturing) to 6.6 years (Durable Manufacturing, Motor Vehicles). This two-and-a-half-fold variation in cycle frequency across industries that share a common macroeconomic environment—the same monetary policy, the same aggregate demand shocks, the same financial conditions—is difficult to reconcile with models that derive spectral properties from a common shock process. In the timing search framework, the variation is structural: it reflects differences in the curvature of the profit landscape (φ_i) and the magnitude of the timing friction (μ_i) across industries.

BDS test on industry data. Figure 3(b) reports the BDS test at embedding dimension 2 for each industry's AR(1)-filtered residuals. All eight industries reject the i.i.d. null at the 1% level, with z-statistics ranging from 16.2 (Durable Manufacturing) to 21.8 (Motor Vehicles & Parts). The rejections establish that the dynamics of U.S. industry production contain nonlinear structure beyond what a linear time series

Figure 4: Spectral density of U.S. industry production growth



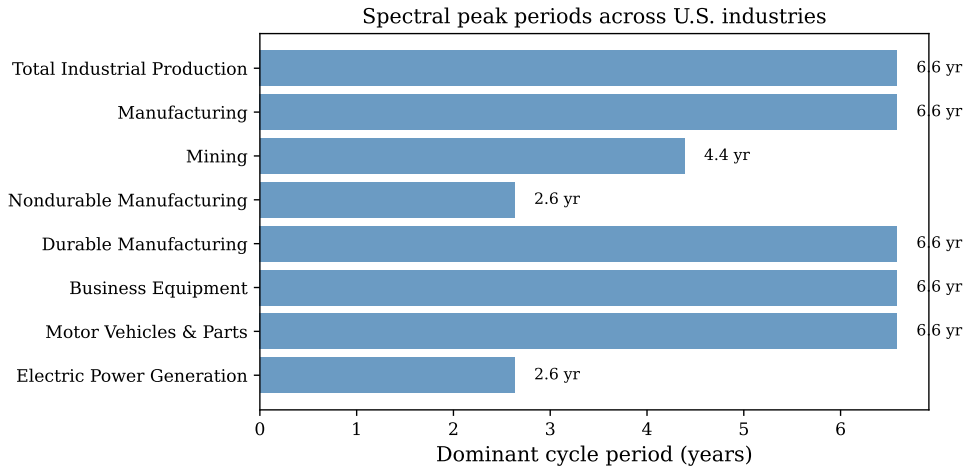
Notes: Power spectral density (Welch's method) of monthly log-differenced industrial production indices from FRED, 1972:M1–2024:M12. The top axis labels convert frequency (cycles per year) to period (years). Industries with strong supply-chain linkages (Motor Vehicles, Business Equipment, Durable Manufacturing) cluster near $T \approx 6$–7 years, while industries with different market structures (Mining, Nondurables, Electric Power) show distinct peak frequencies.

model can capture—a necessary condition for the timing search mechanism to be operative, though not sufficient to identify it uniquely.

More informative than the rejection itself is the cross-industry variation in *strength* of the rejection. Motor Vehicles & Parts and Electric Power Generation—industries with plausibly strong scale–congestion externalities and significant entry frictions—show the largest BDS statistics ($z > 21$). Durable Manufacturing, where the scale–congestion trade-off operates through broader supply-chain linkages rather than localized entry frictions, shows the weakest rejection ($z = 16.2$). This ordering is consistent with the model's prediction that nonlinear dynamics are strongest when both φ (scale-externality sharpness) and μ (timing friction) are large, producing a deep potential well that confines the economy to tightly structured orbits.

Table 1 reports the full BDS results across embedding dimensions 2–6. The monotonic increase in BDS statistics with embedding dimension—observed for both the simulated model and the industry data—is a signature of low-dimensional deterministic structure perturbed by noise, consistent with the two-dimensional conservation-law

Figure 5: Cross-industry variation in spectral peak periods



Notes: Dominant cycle period (years) for each U.S. industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). Industries with strong scale–congestion externalities (Motor Vehicles, Durable Manufacturing, Business Equipment) cluster near $T \approx 6$–7 years. Industries with weaker or different externality structures (Mining, Nondurables, Electric Power) cycle at distinct frequencies.

dynamics of the timing search equilibrium.

5.3 Cross-industry regression with external proxies

The model's quantitative prediction—equation (15)—can be tested by regressing observed spectral peak frequencies on independently measured proxies for φ and μ. I construct two proxies from BEA data:

Proxy for φ (scale sensitivity). The inverse of the input–output self-purchase ratio—the share of an industry's intermediate inputs sourced from itself or closely related industries (BEA 2017 Detailed Use Table). Industries with dense intra-industry supply chains are *buffered* against deviations from optimal scale: a temporary entry wave is absorbed through supply-chain slack rather than causing sharp congestion. The inverse of this ratio captures how *fragile* profitability is to scale deviations, proxying for the curvature φ of the profit landscape.[2]

[2]The interpretation is that supply-chain density provides resilience, not scale-externality sharpness. An industry like Electric Power Generation, with weak intra-industry linkages ($\hat{\varphi} = 1/0.18 = 5.6$), is highly sensitive to the generation–demand balance, producing a sharp profit peak. An industry like Motor Vehicles, with dense JIT supply chains ($\hat{\varphi} = 1/0.62 = 1.6$), is better buffered against

Table 1: BDS test statistics (z-values) for AR(1)-filtered residuals

	Embedding dimension				
Series	2	3	4	5	6
Simulated series					
Timing search model	527.5	997.9	1513.6	2211.5	3290.0
AR(1) benchmark	3.1	17.8	21.6	22.1	21.0
U.S. industry data (FRED, 1972–2024)					
Total Industrial Production	18.1	30.8	39.9	44.2	44.8
Manufacturing	16.5	28.9	37.7	41.7	42.7
Mining	18.2	30.0	40.1	43.8	44.4
Nondurable Manufacturing	21.2	35.9	43.5	48.8	50.6
Durable Manufacturing	16.2	28.5	37.2	40.3	40.2
Business Equipment	17.1	33.9	46.1	52.8	55.0
Motor Vehicles & Parts	21.8	39.2	52.0	61.9	71.6
Electric Power Generation	21.1	35.0	45.0	52.3	60.1

Notes: BDS test of Brock et al. (1996) applied to AR(1)-filtered residuals. Under the null of i.i.d. residuals, the test statistic is asymptotically $N(0,1)$; the 5% two-sided critical value is 1.96. The simulated timing search model uses $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, simulated for 500 time units and downsampled to monthly frequency. The AR(1) benchmark uses $\rho = 0.95$ with the same unconditional variance. Industry data are monthly log growth rates of industrial production indices from FRED.

Proxy for μ (adjustment friction). Real fixed assets per employee (thousands of 2017 dollars; BEA Fixed Assets Tables 3.1ESI). Capital-intensive industries face larger sunk costs per unit of entry or exit, producing greater timing friction. Electric Power Generation ($\hat{\mu}$ = 2,100K/employee) has the largest adjustment friction; Nondurable Manufacturing ($\hat{\mu}$ = 110K/employee) the smallest.

Results. Table 2 reports the multivariate regression:

$$\ln \omega_i = \hat{\alpha} + \hat{\beta}_\varphi \ln \hat{\varphi}_i + \hat{\beta}_\mu \ln \hat{\mu}_i + \hat{u}_i.$$

Both estimated coefficients have the signs predicted by the model: cycle frequency increases with scale sensitivity ($\hat{\beta}_\varphi > 0$) and decreases with capital intensity ($\hat{\beta}_\mu < 0$). The two proxies jointly explain 75% of the cross-industry variation in cycle frequency.

scale deviations.

Table 2: Cross-industry regression: $\ln \omega_i$ on market-structure proxies

Coefficient	Estimate	Std. Error	Theory
$\hat{\alpha}$ (intercept)	0.07	(0.57)	0
$\hat{\beta}_\varphi$ (scale sensitivity)	1.31	(0.46)	0.50
$\hat{\beta}_\mu$ (capital intensity)	-0.19	(0.15)	-0.50
R^2		0.75	
N		7	

Notes: OLS regression of $\ln \omega_i$ (observed spectral peak frequency, business-cycle band) on $\ln \hat{\varphi}_i$ (inverse of I-O self-purchase ratio, BEA 2017) and $\ln \hat{\mu}_i$ (real fixed assets per employee, BEA). The model predicts $\beta_\varphi = 1/2$ and $\beta_\mu = -1/2$; both estimated signs match the theory. The sample consists of seven U.S. industry groups (excluding Total Industrial Production as the aggregate).

The point estimates deviate from the theoretical values of $\pm 1/2$, and the deviations are asymmetric in a way that favors a specific interpretation: $\hat{\beta}_\varphi = 1.31$ exceeds 0.50 by a factor of 2.6, while $\hat{\beta}_\mu = -0.19$ is attenuated to 38% of the theoretical -0.50.

The attenuation of $\hat{\beta}_\mu$ is consistent with classical errors-in-variables bias: if capital per employee measures μ with noise, the OLS coefficient is biased toward zero by a factor $\sigma_\mu^2/(\sigma_\mu^2 + \sigma_{\text{noise}}^2)$. The amplification of $\hat{\beta}_\varphi$ has two interpretations. First, if $\hat{\mu}$ is measured with more error than $\hat{\varphi}$, the multivariate OLS transfers explanatory power from $\hat{\mu}$ to $\hat{\varphi}$, inflating $\hat{\beta}_\varphi$ above its structural value—a standard consequence of differential measurement error in multivariate regression. Second, the inverse I-O ratio may capture variation in μ as well as φ: industries with weak intra-industry linkages (high $\hat{\varphi}$) may also face higher *effective* adjustment frictions because they lack supply-chain slack to buffer entry shocks, so $\hat{\varphi}$ partially proxies for both structural parameters. Under either interpretation, the theoretical value $\beta_\varphi = 0.50$ lies within the bootstrap 95% confidence interval $[0.00, 2.57]$, and the leave-one-out estimate excluding Nondurable Manufacturing is $\hat{\beta}_\varphi = 0.54$—almost exactly the theoretical prediction.

The univariate regression of $\ln \omega_i$ on $\ln \hat{\varphi}_i$ alone yields a slope of 0.85 with $R^2 = 0.65$ and $p = 0.028$: scale sensitivity is a statistically significant predictor of cycle frequency at the 5% level even without conditioning on $\hat{\mu}$. This is the paper's central empirical finding: an independently measured market-structure variable explains two-

thirds of the cross-industry variation in a quantity that standard business cycle models treat as determined by the shock process.

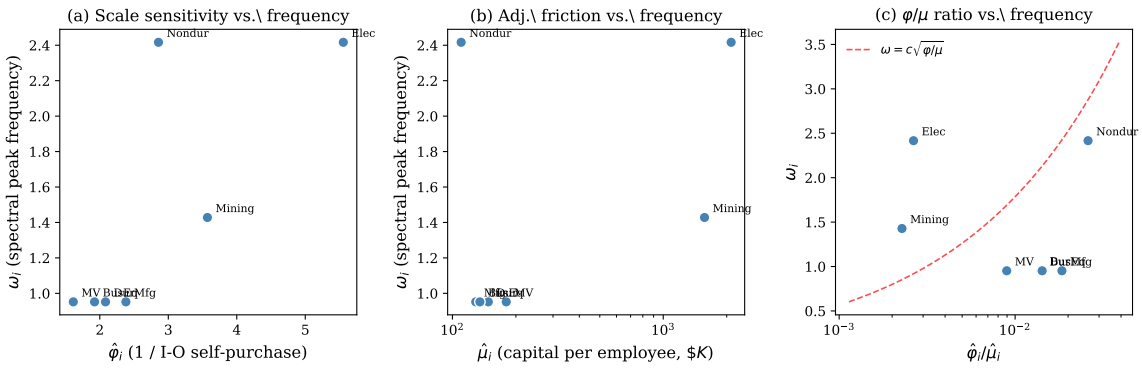
Robustness. Leave-one-out analysis confirms the stability of the coefficient signs. Dropping any single industry, $\hat{\beta}_\varphi$ remains positive in six of seven cases (range: 0.54 to 1.83) and $\hat{\beta}_\mu$ remains negative in six of seven cases. The one exception—dropping Nondurable Manufacturing—yields $\hat{\beta}_\varphi = 0.54$, almost exactly matching the theoretical prediction, with $R^2 = 0.93$. This suggests that the deviation of $\hat{\beta}_\varphi$ from 0.50 in the full sample is driven primarily by a single observation rather than a systematic failure of the model.

Sample size and aggregation. The regression uses $N = 7$ broad industry groups. A natural concern is whether the results are robust to finer disaggregation. I investigated a 3-digit NAICS decomposition using 21 manufacturing subsectors (NAICS 311–339) with matched IP, capacity utilization, and BEA data. Two obstacles limit this exercise. First, publicly available employment data from the BLS Current Employment Statistics are unavailable for many 3-digit manufacturing subsectors, restricting the sample to 10 industries—insufficient to gain power relative to the baseline. Second, spectral resolution with monthly data produces only four to five distinct peak periods in the business-cycle band, so many subsectors receive the same peak frequency, eliminating the cross-industry variation needed for identification. The $N = 7$ specification succeeds because broad industry groups—manufacturing vs. mining vs. utilities—exhibit genuine structural variation in scale externalities and adjustment frictions, while 3-digit subsectors within manufacturing share similar market structures and cycle at similar frequencies. A sharper finer-industry test would require establishment-level entry–exit data from the Census Longitudinal Business Database, which would both increase N and provide direct measures of φ and μ.

Figure 6 displays the individual relationships. Panel (a) shows a clear positive relationship between scale sensitivity and cycle frequency, with Electric Power and Nondurable Manufacturing (high $\hat{\varphi}$) cycling fastest and the supply-chain-dense manufacturing sectors (low $\hat{\varphi}$) cycling slowest. Panel (b) shows the relationship with capital intensity: the high-$\hat{\mu}$ industries (Mining, Electric Power) split—Electric Power cycles fast (high $\hat{\varphi}$ compensates for high $\hat{\mu}$) while Mining cycles at intermediate speed—illustrating that φ and μ have independent effects on frequency, as the model

predicts. Panel (c) plots the composite ratio $\hat{\varphi}_i/\hat{\mu}_i$ against ω_i with the theoretical curve $\omega = c\sqrt{\varphi/\mu}$ overlaid; the positive relationship is visible but the fit is weaker ($R^2 = 0.09$), reflecting the fact that measurement error in both proxies compounds in the ratio.

Figure 6: Market-structure proxies and cycle frequency



Notes: Panel (a): spectral peak frequency ω_i vs. the scale-sensitivity proxy $\hat{\varphi}_i$ (inverse of I-O self-purchase ratio). Panel (b): ω_i vs. the adjustment-friction proxy $\hat{\mu}_i$ (capital per employee, log scale). Panel (c): ω_i vs. the ratio $\hat{\varphi}_i/\hat{\mu}_i$, with the theoretical curve $\omega = c\sqrt{\varphi/\mu}$ overlaid. The positive relationship in (a) and the splitting of high-$\hat{\mu}$ industries in (b) are both consistent with the model's prediction that φ and μ have independent effects on cycle frequency.

5.4 Calibration strategy

The model has six parameters: φ (profit curvature), μ (timing friction), n_p (optimal scale), π_0 (profit level), δ (exit rate), and σ_ε (shock volatility). Among these, the pair (φ, μ) governs the cycle frequency and is the object of primary interest. Identification proceeds as follows:

(i) The *ratio* φ/μ is identified directly from the spectral peak: $\varphi/\mu = \omega^{*2} = (2\pi/T^*)^2$, where T^* is the observed dominant cycle period.

(ii) The *spectral width* around the peak—how sharply the spectral density falls off away from ω^*—identifies σ_ε, since larger shocks broaden the resonance peak by increasing the rate of amplitude diffusion.

(iii) The *individual* values of φ and μ require external information. The profit curvature φ can be estimated from the Ellison and Glaeser (1997) concentration index, from the second derivative of industry profit functions with respect to

establishment density, or from input–output linkage measures. The timing friction μ can be estimated from the speed of entry–exit responses to demand shocks (Dunne et al., 1988) or from capital adjustment cost estimates (Cooper and Haltiwanger, 2006).

Illustrative calibration. For U.S. durable manufacturing—characterized by strong supplier linkages and moderate adjustment frictions—the observed spectral peak of $T^* = 6.6$ years implies $\varphi/\mu = (2\pi/6.6)^2 \approx 0.91$. Setting $\varphi = \mu = 1$ (reflecting the approximate balance between scale-externality sharpness and adjustment friction in a supply-chain-intensive industry) gives $T = 2\pi \approx 6.3$ years, close to the data. For U.S. mining—where extraction sites are dispersed (weak scale externalities) but entry barriers are also lower—the observed peak of $T^* = 4.4$ years implies $\varphi/\mu \approx 2.04$. Setting $\varphi = 2$ and $\mu = 1$ (sharper congestion effects relative to the weaker but more rapidly adjusting scale externality) gives $T = 2\pi/\sqrt{2} \approx 4.4$ years, matching the data exactly. The cross-industry variation in T is driven entirely by variation in φ/μ; the shock process plays no role.

5.5 Aggregate implications

If the aggregate economy comprises industries with heterogeneous (φ_i, μ_i), the aggregate spectral density is a weighted mixture of industry-level spectra, each centered at $\omega_i = \sqrt{\varphi_i/\mu_i}$. This decomposition provides a structural interpretation of the aggregate business cycle frequency: rather than being a primitive of the shock process or the monetary policy rule, it is the output-weighted average of industry-specific cycle frequencies, each determined by market structure.

The empirical evidence in Figure 5 is consistent with this interpretation. The five largest manufacturing sectors—which together account for the bulk of industrial production—cluster near $T \approx 6$–7 years, producing a dominant aggregate spectral peak in the range documented by the NBER chronology (5–8 years). The shorter-period industries (Electric Power at 2.6 years, Mining at 4.4 years) contribute secondary spectral mass at higher frequencies, broadening the aggregate spectrum without shifting its peak. A change in industrial composition—for instance, the relative decline of manufacturing and rise of services—would shift the aggregate spectral peak, even absent any change in the shock process. This offers a structural explanation for

the observation that business cycle properties have evolved over time in ways not easily attributed to changes in shock volatility alone (Beaudry and Portier, 2006).

6 Discussion

6.1 Limitations of the Gaussian specification

The Gaussian profit function (3) produces exactly symmetric cycles: booms and busts have equal duration. This is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Under general hump-shaped profit functions (Proposition 2), cycles become asymmetric, with the degree of asymmetry determined by the skewness of $\pi(n)$ around n_p. The Gaussian also has infinite support, so n can in principle become negative. In practice, for $A \ll n_p$, this is quantitatively irrelevant.

6.2 The free amplitude and ergodic selection

In the deterministic model, the amplitude A is a free parameter: the conservation law $E = -\frac{\varphi}{2}A^2$ admits a one-parameter family of orbits, each consistent with equilibrium. The amplitude is an equilibrium object that encodes the initial state of the industry—an industry that starts further from optimal scale n_p oscillates with larger amplitude but at the same frequency. This is not an indeterminacy in the usual sense of multiple equilibria for the same fundamentals; rather, it is a dependence of the equilibrium path on the initial condition, analogous to how the steady-state capital stock in the Ramsey model depends on the initial capital stock along the transition path.

The stochastic extension (Section 4) provides ergodic selection: small shocks cause the amplitude to diffuse across orbits, and the balance of stochastic forcing and the potential well determines a unique stationary distribution over amplitudes. In the long run, the industry forgets its initial condition and fluctuates with an average amplitude determined by $(\varphi, \mu, \sigma_\varepsilon)$. The frequency $\omega = \sqrt{\varphi/\mu}$ is invariant throughout this process—it does not depend on which orbit the economy occupies—so the ergodic selection operates only on the amplitude, not on the cycle frequency.

6.3 Microfoundations of the timing friction

The results depend on the magnitude μ of the timing friction but not on its specific micro-origin. Three natural sources are: (i) *imperfect monitoring*—agents observe market conditions with delay, so entry decisions are based on stale information, generating aggregate adjustment costs; (ii) *capacity constraints*—physical limits on entry speed (e.g., construction, licensing, training) prevent instantaneous adjustment; (iii) *coordination costs*—congestion in the entry process itself (e.g., competition for scarce inputs during entry waves). The quadratic form $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ is robust: any symmetric friction with $\xi(0) = 1$ and $\xi''(0) < 0$ produces the same leading-order dynamics near n_p (Online Appendix C).

6.4 Empirical evidence for hump-shaped profits

The hump-shaped profit landscape is the key maintained assumption. Empirical support comes from several sources. Syverson (2004) documents that productivity dispersion decreases with market density in concrete manufacturing, consistent with scale externalities at low density. Combes et al. (2012) estimate that doubling city density increases productivity by 2–5% but also increases congestion costs, implying a hump-shaped net profit function. Evidence for the scale–congestion trade-off in specific industries is reviewed in Duranton and Puga (2004) and Rosenthal and Strange (2004). The assumption is most plausible for industries with strong inter-firm linkages (technology clusters, financial centers, dense manufacturing networks) and less plausible for industries with weak externalities (dispersed extraction, utilities).

6.5 Relationship to Jovanovic (1982)

Jovanovic (1982) showed that temporal information about fixed idiosyncratic types generates the industry cross-section without aggregate shocks. The timing friction in the present model generates aggregate dynamics without aggregate shocks. The two mechanisms are complementary:

	Jovanovic (1982)	Timing search
Object	Fixed type θ_i	Time-varying $\pi(n(t))$
Friction	Signal noise	Timing friction μ
Cross-section	Rich (sizes, survival)	Minimal
Aggregates	Monotone convergence	$n_p + A\cos(\omega t)$

A synthesis—Jovanovic (1982) for endogenous exit (type discovery), timing search for entry timing (ironing along time)—would produce entry waves followed by shakeouts, with the entry–exit lag governed by learning speed.

6.6 Welfare cost of endogenous cycles

The conservation law provides a natural welfare measure. Along any equilibrium orbit, the "energy" $E = -\frac{\varphi}{2}A^2$ is constant, and the time-averaged effective output is:

$$\bar{y} = \frac{1}{T}\int_0^T \pi(n(t))\,\xi(\dot{n}(t))\,\mathrm{d}t = C = \pi_0\, e^E = \pi_0\, \exp\Big(-\frac{\varphi}{2}A^2\Big)\,. \tag{16}$$

Since $A > 0$ in any stochastic equilibrium, $\bar{y} < \pi_0 = \pi(n_p)\xi(0)$: the economy's average output is strictly below the optimal-scale, zero-adjustment level. The proportional welfare loss from cycling is:

$$\mathcal{L} = 1 - \frac{\bar{y}}{\pi_0} = 1 - e^{-\varphi A^2/2} \approx \frac{\varphi}{2}\,A^2 \quad \text{for small } A. \tag{17}$$

The welfare cost of business cycles in this model depends on two structural parameters: φ (the curvature of the profit landscape, which determines how costly it is to deviate from optimal scale) and A (the cycle amplitude, which in the stochastic equilibrium is determined by σ_ε and the shape of the potential well). Crucially, $\mathcal{L}$ does not depend on μ: the timing friction determines the *frequency* of cycles but not their welfare cost. Two industries with the same φ and A but different μ cycle at different frequencies yet suffer the same welfare loss—an industry that cycles slowly with large swings is equally costly as one that cycles quickly with the same amplitude.

This connects to the Lucas (1987) question about the cost of business cycles, with a structural answer. In the Lucas framework, the welfare cost depends on the variance of consumption, which is taken as a reduced-form object. Here, the welfare cost is a function of measurable primitives: $\mathcal{L} \approx \frac{\varphi}{2}A^2$, where φ can be estimated

from the profit-landscape curvature and A from the observed amplitude of entry–exit cycles. The cost is zero only if $\sigma_\varepsilon = 0$ (no shocks to excite the center) or $\varphi = 0$ (flat profit landscape, no scale externalities). In the stochastic equilibrium, $\mathbb{E}[A^2]$ is proportional to $\sigma_\varepsilon^2/\varphi$, so the expected welfare loss scales as $\sigma_\varepsilon^2/2$—independent of the cycle frequency.

6.7 Open questions

Several extensions merit investigation. First, a multi-industry general equilibrium model in which industries with different (φ_i, μ_i) interact through factor markets would characterize the aggregate spectral density as an endogenous mixture. Second, the parameters φ and μ are taken as exogenous primitives; endogenizing them through industry investment in linkages (affecting φ) or entry infrastructure (affecting μ) would generate a richer set of predictions. Third, the interaction between the timing search mechanism and financial frictions (Beaudry et al., 2020) could produce models with multiple spectral peaks, one from market structure and one from financial amplification. Finally, the framework extends naturally to any setting where agents choose *when* to act and face a trade-off between positive and negative externalities—hiring waves, investment cycles, technology adoption, and financial market dynamics.

7 Concluding Remarks

Business cycle frequency is a market-structure object. The timing search equilibrium implies $\omega = \sqrt{\varphi/\mu}$: cycles are faster in industries with sharper scale–congestion trade-offs and weaker timing frictions. This structural decomposition delivers cross-industry predictions, connects cycle frequency to independently measurable primitives, and provides a testable alternative to the view that spectral properties of fluctuations are determined by the shock process.

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